

Assessment Task No. 1 - Introduction to Machine Learning

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BSCS - 3B

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CSST102 – Basic Machine Learning

From Clicks to Cart: Designing a Machine Learning-Based Product Recommendation System

Chosen Machine Learning Application:

Product Recommendation - A system that gives recommendations for products to be bought by the user during online shopping, just like Shopee, Lazada, Tiktok shop does.

Problem Description:

There are thousands and even millions of products available on online shopping websites. Therefore, it becomes difficult for the customer to browse and find out the required product. This problem is solved by the product recommendation system through which the activities of the customer on the website such as viewing, purchase, placing orders in the shopping cart, and clicking are considered to provide recommendations of related items. Through this method, the customer is able to find out the relevant item quickly, and it is beneficial for the business as well since customized recommendations result in better sales.

Part A - Identify the Machine Learning Problem

Questions	Your Answer
1. What is the application?	Basically, the application we see in online shopping apps such as Shopee, Lazada, Tiktok Shop, Shein, and many others.
2. What problem are you trying to solve?	Basically, recommending products to the customer based on his/her browsing behavior rather than recommending products to all customers.
3. What are the inputs?	Some of these variables might consist of the browsing record of the consumer, prior purchases done by the person, the items put in the shopping cart, search results, and ratings.
4. What is the expected output?	The customer's desired items which the system is predicting to be displayed in a list.
5. What patterns or relationships might the ML system learn?	Perhaps the idea that consumers who purchase or pick up items from a particular category will generally tend to buy additional items from that category or a similar one.
6. What data would the system need?	Information relating to product identification numbers, categories, information related to customer browsing history, purchasing times, and rating.

Part B - Traditional Programming vs. Machine Learning

Traditional Programming (Input → Algorithm → Output)

In traditional programming, the programmer needs to know beforehand about the rules and write them manually, such as “If the customer has bought a phone, then recommend him the phone case,” or “If the customer has bought a bike, then recommend him the bike parts.” In terms of the recommendation system for the products, the programmer would need to create rules for almost all possible combinations of the products, which would be practically impossible due to the sheer number of products and constant changing behaviors of customers.

Machine Learning (Input + Output/Data → ML Algorithm)

In Machine Learning, the process is completely different. The programmer does not define the recommendation rules. Rather, the algorithm is fed large amounts of data on how the customer acts together with what he ended up buying, and it deduces the pattern from that. This allows for much greater scalability because the model is capable of changing its recommendation rules autonomously in response to changes in products, customers, or trends.

Part C - Create your own Mini Dataset

Below is a simple dataset with 5 observations showing the relationship between how many items a customer bought from a category and how many similar items the system recommends afterward.

Input (No. of items bought in the same category)	Output (No. of similar products recommended)
1 (Ex: T-shirt for men)	3 (Polo Shirt, Perfume, Pants, Shorts)
2	6
3	9
4	12
5	15

Pattern/Relationship:

In other words, the pattern I noticed here is that the number of product suggestions by the system goes up by three with each additional purchase made by the customer within that particular category. For instance, if the customer buys one item, then there will be three suggestions. However, since there has been no modification to the sequence, the number that has been missed out is 15, considering that the input number is 5. It would then be evident once the sequence has been put into the equation format that the output is 3 times the input.

Now, in terms of how the algorithm would be able to detect this very same pattern, I believe that, basically, it analyzes the pairs of inputs and outputs that we already know such as 1→3, 2→6, 3→9, 4→12 and tries to make various hypotheses as to what is going on there. With each hypothesis, the system evaluates how accurate its output is whether it is similar or different from what it should be. The algorithm then gradually improves itself until the difference between what the machine predicts and what it should predict becomes really small. And once the system analyzed enough examples, it will understand that the pattern simply is “multiplying the input

number by 3,” and therefore, if it gets any new number say, 5 it will immediately guess that the result should be 15.

Part D – Explain the Training Process

How does a Machine Learning algorithm learn from data?

First of all, the algorithm receives some inputs such as customer browsing and purchase history; in addition, it is shown the right output that includes purchases made or links visited by the customer in the future. Throughout the learning period, the model constantly makes predictions and measures the accuracy of its predictions in regard to real results. Every time the model makes a prediction that is not quite accurate, it makes some adjustments so that the next time it will do better. The model does this many times in order to gradually learn the right relationship between inputs and outputs. Once the process of learning is completed, the model is ready to predict the output based on some new inputs that it has never seen before.

Short Reflection:

Why should we learn from data when we can program all the rules manually?

Learning from data is better than programming everything by hand since it's basically impossible for a human being to formulate rules for every single occasion, especially when there are hundreds or thousands of goods and customers in the shop. Customers' tastes change all the time too, so there is no way to formulate an up-to-date set of rules for them. ML bypasses the problem by allowing the computer to find those patterns on its own instead of trying to formulate them manually.